Setup

For this program, I was developing with a MacBook pro that used the M1 pro chip. I had to follow the installation process that met the criteria of my architecture. For this I had to install Rosetta 2 so it would translate the intel-based code. To do this I opened the terminal and ran the command to install it that I found on the installation guide. Then the next file I had to download was XQuartz. This was because Wine would use the X11 windowing system to draw the game window onto my mac but MacOS doesn't natively support this anymore. So similarly, I installed it directly from my terminal with the command found on the guide. The last file for the setup was Wine which allows window applications (like the .exe file for torcs) on MacOS to run. I installed wine using Homebrew from the terminal using the designated command. After wine was installed, it created a windows C: drive on my Mac so it could now store windows system files. Now as the setup was complete, I installed the TORCS folder from the download link onto my system and saved it onto the new C: drive that was created. To run the application, I used the following commands in my terminal "cd ~/.wine/drive_c/torcs/torcs" and "wine wtorcs.exe". This would open the application where I could begin the set up for TORCS. I began to configure the race by navigating through the menu options race > quick race > configure race where I selected the oval track "A-Speedway" then selected scr_server 1 as my racer by clicking on it and pressing accept. Then I configured the laps to 1 just to practice getting the car around the track. After it was configured I would enter new race and opened a second terminal where I would enter the command "cd ~/.wine/drive_c/torcs/gym_torcs/" to navigate to the folder where the python scripts were located and "python3 torcs_jm_par.py" which was the main script that I would use to get the car moving around the track.

Coding

When I ran the "python3 torcs_jm_par.py" from the terminal I noticed that the original code left the car to oscillate frequently and it would occasionally crash into the wall and spin out. It was unable to complete a single lap around the track. To begin fixing this I looked at the current steering logic. I noticed it was way to aggressive in the sense that the multiplier (25 / PI) meant that if there was a tiny error in the angle, the car would begin to steer really hard. This led the car to keep drifting left and right until it spun out. To fix this I set the steer gain from ~8.0 to 3.0 making the sensitivity a lot lower. This led the car to softly return to the centre rather than constantly trying to fix itself. This later on allowed me to move to higher speeds as it was more stable. The next issue that I addressed was the throttling. In the original code the throttle would instantly add 40% gas. This made it hard to slow down for corners so it would occasionally crash due to it understeering. To address this, I made a check to see if the steering wheel was straight and if so it would aim for the target speed. But if the car was doing a hard turn, the target speed would drop automatically by a simple subtraction so the car was kept on the track during turns. I initially set the speed to 90, the steer gain to 3.0, the centering gain to 0.15, and the brake threshold to 0.4. I dropped the speed from 160 to 90 as I wanted to initially get the car moving and finish a single lap around the track as it made it easier to analyse what other factor beside speed was causing an issue. The steering and centering were significantly lowered to ensure stability and prevent the crashes and harsh adjustments the car previously made. The brake threshold was quite loose as it would allow the car to retain momentum through the corners and only break if it was spinning out. In the end the car was able to complete a single lap through the circuit without crashing into any of the other cars, spinning out or nose diving into the barriers / walls.